SimpsonEF: EXPLAINABLE EJECTION FRACTION FROM PREDICTED SONOGRAPHER TRACINGS AND SIMPSON'S RULE

Gowthamaan Palani, Ganapathy Krishnamurthi

Department of Data Science and Artificial Intelligence
Indian Institute of Technology Madras, Chennai, India

ABSTRACT

Video networks estimate left-ventricular ejection fraction (EF) well but return a number whose derivation cannot be checked. SimpsonEF computes EF the way a sonographer does: it predicts the volume tracing (long axis and 20 chords) at end-diastole (ED) and end-systole (ES) of every beat and applies Simpson's method of disks, so the reported EF is a fixed formula of drawn, inspectable geometry. Three small networks trace every frame (FrameTracer), find the beats on the resulting volume curve (BeatFinder) and jointly redraw each beat's ED and ES tracings (BeatTracer), the last trained through Simpson's formula in EF units. On all 1277 EchoNet-Dynamic test videos, with no annotation at inference, SimpsonEF reaches R^2 0.806 and MAE 4.01 EF points with 4.5M parameters: non-inferior to the video regressors we re-ran on the same videos, and 0.16 above previously reported keypoint-derived EF. Because the output is a formula, it can adopt a new dataset's measurement convention: zero-shot on CAMUS, whose label is a biplane Simpson EF, it reaches R^2 0.710 (95% CI 0.49–0.84), where eight released EF models reach at most 0.28. Conformal intervals cover 90.1% of test cases, and the tracings segment the ventricle with Dice 0.935/0.915.

Index Terms— Ejection fraction, echocardiography, Simpson's method, keypoints, explainability, conformal prediction

1. INTRODUCTION

Left-ventricular ejection fraction (LVEF, here EF), the fraction of the end-diastolic volume ejected in a beat, is the principal measure of systolic function [1]. Video networks estimate EF to about 4 EF points [2, 3] but return a bare number. Clinicians compute EF from a tracing by Simpson's method of disks, biplane where possible [4], so a model whose output is that tracing, and whose EF is that formula, can be checked frame by frame.

Related work. Direct regression with 3D CNNs [2], CNN–transformers [5], contrastive regression [3] or light spatio-temporal networks [6] reaches R^2 0.81–0.82. Foundation models pretrained on millions of echocardiograms [7, 8, 9, 10] give EF estimates but are black boxes with tens of millions to 1.1B parameters. Explainable models use graph attention [11] or prototypes [12] but still regress EF; uncertainty-aware ones use Bayesian networks [13] or conformal prediction [14]. Segmentation pipelines derive EF from predicted masks [15], although explicit masks do not improve learned EF regression [16]. Keypoint models come closest to the clinical computation: EchoTrace [17] regresses the 84 tracing coordinates of single frames, and EchoGraphs [18] finds contour keypoints with a graph network but reports that Simpson's EF from them (R^2 0.65 on whole videos) trails a black-box regressor (0.79).

We contribute: (i) a video-to-EF pipeline whose output is a fixed Simpson computation on predicted tracings, with every intermediate inspectable; (ii) joint ED/ES tracing trained through the formula in EF units, which closes the MAE gap to direct regression without degrading the anatomy; (iii) a like-for-like evaluation against eight released models on EchoNet-Dynamic and zero-shot on CAMUS, covering EF, segmentation, ED/ES detection and uncertainty; and (iv) an analysis of where formula EF loses accuracy.

2. METHOD

Tracings and EF. A tracing has the ends A_1, A_2 of the long axis and the ends L_i, R_i of 20 perpendicular chords (EchoNet-Dynamic convention). With $d_i = \|L_i - R_i\|$ and $\ell = \|A_1 - A_2\|$,

$$V = \frac{\pi}{4}\,\frac{\ell}{20}\sum_{i=1}^{20} d_i^2, \qquad \mathrm{EF} = 100\Big(1 - \frac{V_{\mathrm{ES}}}{V_{\mathrm{ED}}}\Big). \tag{1}$$

No mask is a training target or enters the EF; masks only measure Dice.

Tracers. FrameTracer and BeatTracer are fully convolutional encoder–decoders (instance norm, GELU, skip connections, dilated bottleneck, no global pooling) with one full-resolution heatmap per point; each point is the soft-argmax [19] over a 7×7 window at the peak. The same weights run at scales 1, 1/2 and 1/4 and the logits are fused by log-sum-exp, so FrameTracer's VAL Dice changes by at most 0.011 between 0.5× and 3× the training size. FrameTracer (1.42M parameters) takes 13 frames (offsets −24..24, step 4) and returns 42 points. BeatTracer (2.53M, width 32) takes the same 13-frame windows around ED and around ES stacked as channels and returns both tracings (84 points), so ES is drawn knowing the ED appearance. Chords are predicted as free points, not constrained: on TEST they lie within 1.6° of perpendicular to the axis (spacing irregularity 7%).

Beats. BeatFinder, a two-layer bidirectional GRU (0.53M), reads FrameTracer's chord sequence $x_t = (d_1, \ldots, d_{20}, \ell)$ and outputs ED and ES heatmaps over time (Gaussian targets around the annotated beat); peaks give each beat's ED and following ES. It only chooses frames, and its choices can be checked on the volume curve.

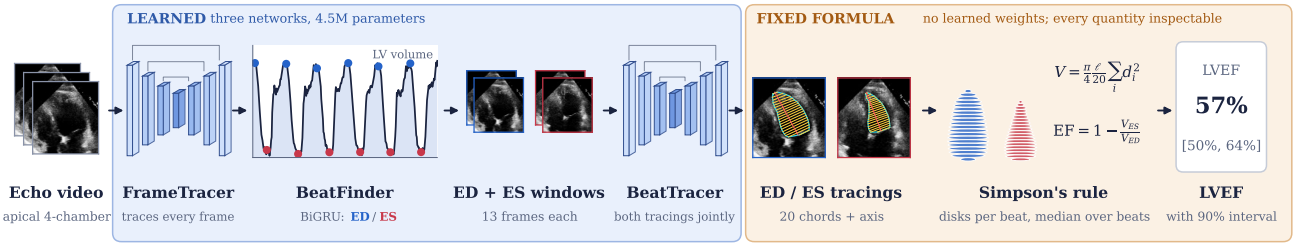


Fig. 1. SimpsonEF on a TEST case. Learned (blue): FrameTracer traces every frame, BeatFinder marks each beat's ED and ES on the volume curve, BeatTracer redraws both tracings from the ED and ES windows. Fixed (orange): Simpson's rule turns the chords into volumes; LVEF is the median over beats, with a 90% interval.

Training through the formula. Both tracers use an L_1 point loss, heatmap cross-entropy (σ=1.5 px) and L_1 losses on $\log V$, on its ED→ES change and on the log chord widths weighted by their share of V, with similarity transforms (rotation ±20°, scale 0.8–1.2, shift ±18 px, frame rescaled 0.5–2.5× on its canvas), edge occlusion and scanner-style degradations (gain, gamma, speckle, noise, blur, shadows). The change of $\log V$ is $\log(1-\mathrm{EF})$, so an EF point costs $\approx 1/(1-\mathrm{EF})$: 1.4 at EF 30% but 3.3 at 70%, i.e. this loss values low-EF hearts least. BeatTracer is therefore fine-tuned (12 epochs, learning rate 10^{-4}, no dropout) with $\mathscr{L}_{\mathrm{EF}} = w_b\,|\mathrm{EF}(\hat{P}_{\mathrm{ED}}, \hat{P}_{\mathrm{ES}}) - \mathrm{EF}^*|$ in EF units, w_b the inverse square root of the frequency of the label's 10-point EF bin. BeatTracer is trained without the scanner-style degradations, since input noise shrinks outputs towards the mean (attenuation bias [20]).

Inference and uncertainty. BeatTracer is applied to 15 rescaled (0.85–1.17) and rotated (±8°) copies of each beat's input and the mapped-back points are averaged; (1) gives the beat's EF and the median over beats is reported. A normalised split-conformal interval [21, 22, 23] uses the scale $s = (\hat{q}_{.95} - \hat{q}_{.05})/2 + |\widehat{\mathrm{EF}} - \widehat{\mathrm{EF}}_{\mathrm{GRU}}|$, where BeatFinder also predicts EF quantiles from FrameTracer's chords: s grows when the clip is uncertain or the two routes to EF disagree. With Q the conformal quantile of $|y - \widehat{\mathrm{EF}}|/s$ on VAL (n=1288), $\widehat{\mathrm{EF}} \pm Qs$ covers 90% of exchangeable cases.

Interpretability. The reported EF is computed only by (1); BeatFinder's learned EF estimate only widens the interval. Every estimate decomposes into quantities a clinician can inspect and correct: the volume curve and chosen frames, the tracings, the 20 chord widths and the axis, the EF of every beat and their median, and the reason for a wide interval.

3. EXPERIMENTS

Protocol. EchoNet-Dynamic [2] has 7465/1288/1277 TRAIN/VAL/TEST videos, each with an EF label and, except five TRAIN videos and one TEST video, one annotated beat (ED/ES frames and tracings). Training uses the 7460 annotated TRAIN videos (minus 81 records with several tracings stacked on one frame; frame windows are clamped at clip boundaries); EF, which needs only the video and its label, is scored on all 1277 TEST videos, Dice and ED/ES detection on the 1276 annotated ones. The final configuration was selected on VAL: the tracer variant by a rule fixed before scoring (highest full-pipeline VAL R^2 among 25 trained variants), then input-copy averaging and aggregation over beats; TEST was run once on the frozen pipeline. VAL also calibrates the intervals, so it serves both selection and calibration. At test time no annotation is used. We report R^2, MAE (EF points) and Pearson r against the EF label, and Dice between the mask drawn from a tracing and the released mask. The tracers are trained from scratch (AdamW, learning rate 3×10^{-4}, cosine decay, EMA 0.999, 32 frames per batch, 40 epochs, then BeatTracer's EF stage) and BeatFinder for 16 epochs; one script rebuilds all three networks with fixed seeds; rebuilding BeatTracer from scratch with another seed gives R^2 0.818 on VAL and 0.797 on TEST (0.823 and 0.806 here). Inference takes 0.49 s per 183-frame video on one RTX 3090 (0.36 s without input-copy averaging).

Comparison with released models. We ran every model whose weights are public (R in the tables) on the same 1277 EchoNet-Dynamic TEST videos and, unchanged, on the 50 CAMUS test patients; models without public weights are listed with the authors' figures (P). Each model uses its official preprocessing, with all clips of a video averaged; foundation models are used zero-shot or as a frozen encoder with a head fitted on EchoNet TRAIN (ridge for EchoFM, its attentive probe for EchoJEPA).

EF. Formula EF reaches R^2 0.806 (95% CI 0.779–0.829), MAE 4.01 and r 0.899 (Table 1), 0.16 above the keypoint-derived EF reported by EchoGraphs on whole videos. Against the EchoNet-trained regressors on the same videos, the paired bootstrap MAE difference is −0.01 (−0.18, +0.16) for EchoNet-Dynamic and +0.09 (−0.08, +0.25) for Echo-E^{3}Net: SimpsonEF is non-inferior at a 0.3-point margin, not better; it beats EchoGNN (−0.45) and the foundation models (0.63 at best). Reduced EF is detected with AUROC 0.978 at EF <40% (EchoNet-Dynamic and Echo-E^{3}Net 0.982; sensitivity/specificity 71.9/98.4% vs 70.6/98.1% and 78.8/98.2%) and 0.956 at <50%. MAE 4.0 is smaller than the 7.6–13.9-point inter-observer variation cited in [2].

Segmentation. Against EchoNet's own masks, BeatTracer's tracings reach Dice 0.935/0.915 (ED/ES) against 0.920/0.897 for the released DeepLabV3 [24, 2] (Table 2, Fig. 2); against the released PNG masks, which include the long-axis ends, 0.917/0.907 against 0.903/0.887. The published specialised segmenters, trained on masks, are at 0.936–0.940 (ED) and 0.916–0.923 (ES), with 7–56M parameters against BeatTracer's 2.5M; they use other evaluation code, and our output shares the chord representation the reference masks are drawn from, so we claim parity, not superiority.

ED/ES detection. Unlike dedicated phase detectors [25, 26], BeatFinder needs no phase labels beyond the one annotated beat; it places ED and ES within 1.79 frames on aver-

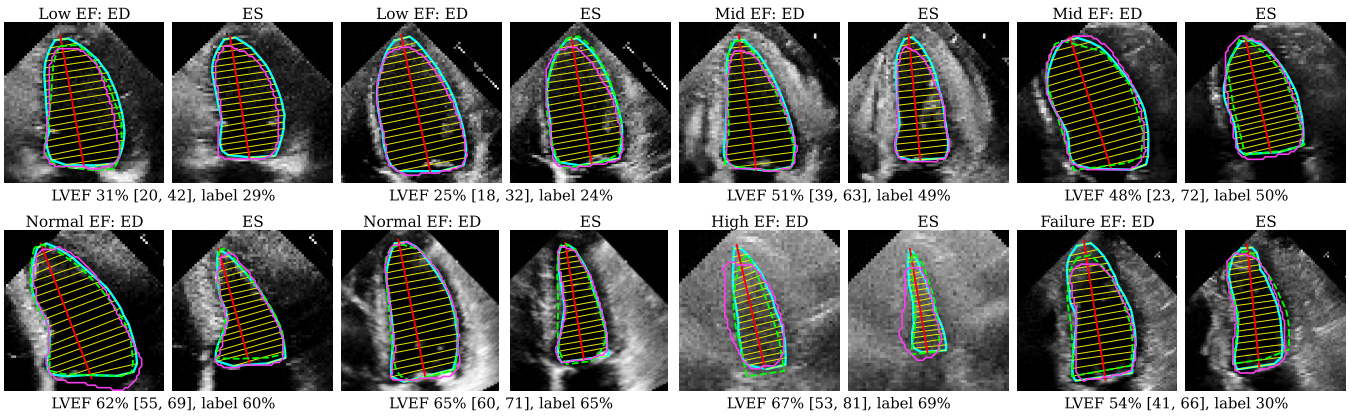


Fig. 2. TEST cases, ED left and ES right, cropped to the tracing: BeatTracer's tracing (cyan, chords yellow, axis red), the released DeepLabV3 mask (magenta), the reference (green dashed); formula EF with its 90% interval and the label. Two each of low, mid-range and normal EF, one high EF, one failure (label outside the interval).

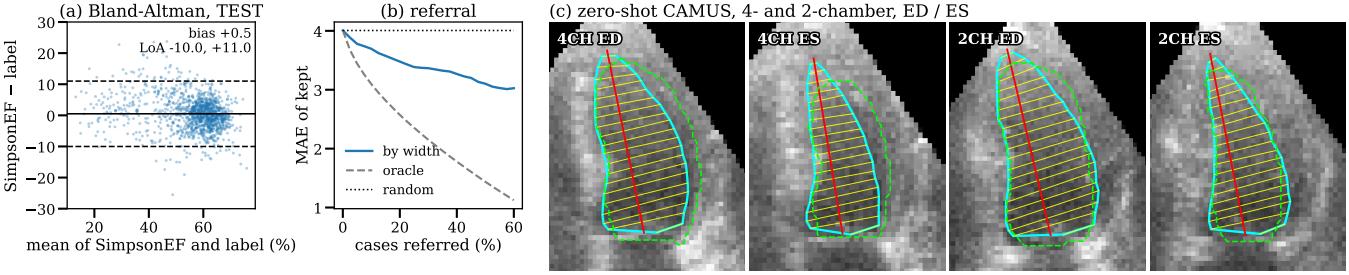


Fig. 3. (a) TEST: Bland–Altman of SimpsonEF against the label. (b) MAE after excluding the widest intervals. (c) Zero-shot CAMUS: FrameTracer's tracing (cyan) and the CAMUS reference (green dashed).

age, 76.5% of them within 2 frames (Table 4); the extremes of FrameTracer's volume curve alone give 3.95 (56.5%), so BeatFinder adds value; EchoGNN's attention gives 3.43 on the videos where it is determinable. On CAMUS half-sequences the error is 2.88 frames (43% within 2).

Training objective. Table 5 shows the steps on VAL (differences below ≈0.01 are within seed variation); the EF-unit formula loss is the largest gain, and Simpson's formula on BeatTracer's tracings outperforms a learned EF head on the same pipeline's chords (0.802 vs 0.823). The loss does not buy EF with worse anatomy: on VAL, Dice (0.916/0.906 before, 0.914/0.905 after) and absolute volume errors (ED 13.2% to 12.7%, ES 16.2% to 15.4%) hold or improve. ED and ES volume errors are correlated (0.79 before, 0.80 after): a shared scale error, present without the EF loss, that cancels in the ratio.

Where the error is. Simpson's rule on the sonographer's own TEST tracings reproduces the label (R^2 0.991 on the 1264 videos with one tracing per frame; 0.943 with the 12 whose tracings are stacked), so the error is in the predicted tracing, and there in the chord widths: at the annotated frames BeatTracer's tracings give 0.790, with the reference widths 0.969, with the reference axis only 0.805. The residual is EF compressed towards normal (slope 0.81; Bland–Altman [34] bias +0.5, limits −10.0 to +11.0, Fig. 3a): +3.4 to +5.7 points below EF 50%, +0.8 to −1.3 in 50–70%, −5.8 above 70%. Tail oversampling flattens it (slope 0.88), but neither it nor width-targeted losses, sharper heatmaps or twice-longer pretraining raise VAL R^2 above 0.822. At the annotated frames, replacing both tracings by the reference recovers the label (MAE 0.84), but replacing only the worse one does not help (4.03 to 4.69), because the shared error then no longer cancels. Of the 89 TEST errors above 10 points, 57 persist at the annotated frames and only 8 stem from beat selection; in the failure of Fig. 2 the ED tracing is 19% too large and the ES 18% too small, errors that do not cancel.

Uncertainty. Calibrated on VAL, the intervals cover 90.1% of TEST cases (95% CI 88.3–91.6) with mean width 16.6 (±8 points, wide clinically), and 87.5% of cases below EF 40% (constant width: 73.1%). Width tracks error: it detects errors above 10 points with AUROC 0.73, and excluding the 20% widest intervals lowers the MAE of the rest from 4.01 to 3.48 (Fig. 3b). VAL also selected the model, so the guarantee is approximate; TEST coverage confirms it.

Zero-shot CAMUS. CAMUS [35] differs in protocol, not only images: (i) its EF label is biplane Simpson (4- and 2-chamber views) with one disk height shared by ED and ES, whereas EchoNet labels are single-plane with per-phase heights; (ii) each view is a half cycle ED→ES (≈20 frames), not a multi-beat video; (iii) contours follow another convention. Frames are padded to a square and resized to 112. We run FrameTracer and BeatFinder unchanged on the 50 test patients, with no CAMUS data in training and no annotated frame, and apply the CAMUS label definition to the chords (its reference code reproduces the label from the reference masks, R^2 0.998). Which tracer to use was decided on the 400 CAMUS training patients, whose labels serve only this choice: biplane R^2 0.55 for FrameTracer, 0.51 for BeatTracer.

Model	Src	Par. (M)	Training data	MAE	R^2
SimpsonEF (ours)	R	4.48	EchoNet	4.01	0.806
EchoNet-Dynamic [2]	R	31.3	EchoNet	4.01	0.815
Echo-E^{3}Net [6]	R	1.76	EchoNet	3.92	0.819
EchoGNN [11]	R	1.7	EchoNet	4.46	0.758
EchoCoTr-B [5]	P	21	EchoNet	3.98	0.81
CoReEcho [3]	P	21	EchoNet	3.90	0.82
EchoGraphs [18]	P	27.6	EchoNet	4.23	0.79
EchoGraphs, Simpson [18]	P	27.6	EchoNet	4.30	0.65
PanEcho [8]	R	42.0	Yale (0-shot)	5.47	0.600
EchoPrime [7]	R	123.1[a]	Cedars (0-shot)	7.78	0.210
EchoCLIP [27]	R	151.5	Cedars (0-shot)	11.51	−0.387
EchoFM-L [28] (probe)	R	344.5[b]	290K clips, probe	5.64	0.627
EchoJEPA-L [10] (probe)	R	303.9	18M echos, probe	5.48	0.630

Table 1. EF on all 1277 EchoNet-Dynamic TEST videos. R: released weights run by us; P: published figures. Par.: parameters (M). Foundation models are zero-shot (0-shot) or a frozen encoder with a head fitted on EchoNet TRAIN only (probe). SimpsonEF: R^2 95% CI 0.779–0.829, MAE 3.81–4.21. [a]Encoder and view classifier. [b]Full masked-autoencoder checkpoint.

Model	Src	Par.	ED	ES	Pool
SimpsonEF (ours)	R	2.53	93.5	91.5	92.5
DeepLabV3 [2]	R	39.6	92.0	89.7	90.9
nnU-Net [29]	P	7.37	93.6	91.6	92.9
SepXception [30]	P	55.8	93.6	91.7	92.9
UniLVSeg [31]	P	18.8	94.0	92.3	93.3
MSV-Mamba [32]	P	n.r.	93.7	91.7	92.9

Table 2. LV segmentation on EchoNet-Dynamic TEST: Dice (%) at the annotated ED and ES frames (1276 annotated videos). R: ours, against EchoNet's own masks (40 chord points); against the released PNG masks SimpsonEF scores 91.6/90.7 and DeepLabV3 90.3/88.7. P: authors' figures.

On the 50 test patients biplane formula EF reaches R^2 0.710 (95% CI 0.49–0.84; MAE 5.08, r 0.849; 4-chamber alone 0.567, 2-chamber alone 0.592; Dice 0.905/0.868 4CH, 0.893/0.864 2CH); with EchoNet's per-phase disk heights the same chords give $R^2 \approx 0$ (bias +9), so the label convention matters as much as the model. The rest is contour convention (Fig. 3c): our contours are narrower towards the base, most at ES; replacing only our 4-chamber tracings by the reference gives 0.79. A zero-shot model cannot learn this convention.

CAMUS against released models. The same models, run unchanged on CAMUS (Table 3), reach R^2 0.28 at best (PanEcho) and the EchoNet-trained regressors 0.08 or less, against 0.710 for SimpsonEF. Like for like on the 4-chamber view alone, SimpsonEF reaches 0.57 (0.18–0.78) against at most 0.22 (−0.26–0.49). The comparison favours a formula: the CAMUS label is itself a biplane Simpson EF, which a regressor cannot know, and CAMUS gives half a cycle per view, so the baselines see a synthetic ED→ES→ED cycle repeated.

Table 5. Ablations on VAL (1288 videos, full pipeline, beats from BeatFinder); the final configuration was selected on these numbers. TEST, run once: R^2 0.806. *25-frame family (its reference: 0.813).

Variant	R^2	MAE	r	slope
FrameTracer tracings only	0.774	4.34	0.881	0.80
+ joint ED/ES tracing (BeatTracer)	0.798	4.08	0.895	0.78
+ EF-unit formula loss	0.817	3.93	0.906	0.82
+ width 32	0.822	3.88	0.907	0.84
+ input-copy averaging (SimpsonEF)	0.823	3.85	0.908	0.83
One change to SimpsonEF				
learned EF head instead of Simpson	0.802	3.96	0.895	0.81
mean instead of median over beats	0.791	4.06	0.890	–
no EF-unit loss	0.811	4.00	0.901	0.80
25-frame windows, 3 EF stages	0.815	3.91	0.904	0.82
tail-oversampled EF stage*	0.802	4.15	0.899	0.88

(a) EF

Model	MAE	R^2
SimpsonEF (ours)	5.08	0.710
EchoNet-Dynamic	9.71	0.079
Echo-E^{3}Net	11.21	−0.303
EchoGNN	9.94	−0.178
PanEcho	7.53	0.283
EchoPrime	18.63	−1.832
EchoCLIP	9.33	0.072
EchoFM-L (probe)	9.10	0.100
EchoJEPA-L (probe)	10.57	−0.099

(b) LV segmentation, Dice (%)

Model	ED	ES	Pool
SimpsonEF (ours)	90.5	86.8	88.7
DeepLabV3	88.1	79.9	84.0

Table 3. Zero-shot CAMUS test (50 patients). (a) EF against the biplane label (ours: biplane Simpson; others: mean of their 4- and 2-chamber EF). (b) 4-chamber LV Dice (%).

Model	Src	Par.	ED	ES	Pool	Acc
SimpsonEF (BeatFinder)	R	1.95	1.76	1.82	1.79	76.5
EchoGNN [11][c]	R	1.7	2.75	4.15	3.43	44.0
Volume-curve extremes[e]	R	1.42	5.22	2.67	3.95	56.5
Latent motion [33][d]	P	n.r.	2.9	2.0	2.4	n.r.

Table 4. ED/ES detection on the 1276 annotated TEST videos: frame error and % within 2 frames (Acc). [c]Videos where determinable (87/83%). [d]Unsupervised. [e]FrameTracer's volume-curve extremes.

4. DISCUSSION AND CONCLUSION

EF computed only by Simpson's rule on predicted tracings is non-inferior to black-box video regressors on EchoNet-Dynamic (paired MAE differences within ±0.3), with 4.5M parameters and every intermediate open to inspection; and because the formula is explicit, it can adopt a new dataset's measurement convention, which a regressor cannot (CAMUS). Limitations: EF stays compressed towards normal by ≈5 points at the extremes; ED/ES supervision is one beat per video; 50 CAMUS test patients give wide intervals; conformal coverage under domain shift is not guaranteed; no clinician correction study was run. Next: biplane EF and a reader study. Code: github.com/gowthamaanp/simpsonef.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study used open-access human subject data from EchoNet-Dynamic (Stanford University) and CAMUS (University Hospital of St Etienne); ethical approval was not required as confirmed by the licences attached to the data.

6. ACKNOWLEDGMENTS

No funding was received. The authors declare no competing interests. AI tools (Claude, Anthropic) assisted with code, analysis, figures and editing; the authors take responsibility for all content.

7. REFERENCES

[1] B. Bozkurt et al., "Universal definition and classification of heart failure," European Journal of Heart Failure, vol. 23, no. 3, pp. 352–380, 2021.

[2] D. Ouyang et al., "Video-based AI for beat-to-beat assessment of cardiac function," Nature, vol. 580, pp. 252–256, 2020.

[3] F. A. Maani et al., "CoReEcho: continuous representation learning for 2D+time echocardiography analysis," in MICCAI, 2024, pp. 591–601.

[4] Roberto M. Lang et al., "Recommendations for cardiac chamber quantification by echocardiography in adults: an update from the American Society of Echocardiography and the European Association of Cardiovascular Imaging," J. Am. Soc. Echocardiogr., vol. 28, no. 1, pp. 1–39, 2015.

[5] Rand Muhtaseb and Mohammad Yaqub, "EchoCoTr: Estimation of the left ventricular ejection fraction from spatiotemporal echocardiography," in Medical Image Computing and Computer Assisted Intervention – MICCAI 2022, 2022, pp. 370–379.

[6] M. Heidari et al., "Echo-E^{3}Net: efficient endocardial spatio-temporal network for ejection fraction estimation," arXiv:2503.17543, 2025.

[7] Milos Vukadinovic et al., "EchoPrime: A multi-video view-informed vision-language model for comprehensive echocardiography interpretation," 2024, arXiv:2410.09704.

[8] Gregory Holste et al., "Complete AI-enabled echocardiography interpretation with multitask deep learning," JAMA, 2025.

[9] Ziyang Zhang et al., "Echo-Vision-FM: a pre-training and fine-tuning framework for echocardiogram video vision foundation model," Nature Communications, 2025.

[10] Alif Munim et al., "EchoJEPA: A latent predictive foundation model for echocardiography," 2026, arXiv:2602.02603.

[11] M. Mokhtari et al., "EchoGNN: explainable ejection fraction estimation with graph neural networks," in MICCAI, 2022, pp. 360–369.

[12] Y. Ghamary et al., "ProtoEFNet: dynamic prototype learning for inherently interpretable ejection fraction estimation in echocardiography," arXiv:2512.03339, 2025.

[13] M. M. Kazemi Esfeh et al., "A deep Bayesian video analysis framework: towards a more robust estimation of ejection fraction," in MICCAI, 2020, pp. 582–590.

[14] P. Fischer et al., "CUTE-MRI: conformalized uncertainty-based framework for time-adaptive MRI," arXiv:2508.14952, 2025.

[15] Hongrong Wei et al., "Temporal-consistent segmentation of echocardiography with co-learning from appearance and shape," in MICCAI, 2020.

[16] F. F. Khouzani et al., "The segmentation ceiling: why explicit left-ventricular masks do not improve learned ejection-fraction regression," arXiv:2609.19730, 2026.

[17] A. K. Paul and Y. S. Bhuiyan, "EchoTrace: a 2D echocardiography deep learning approach for left ventricular ejection fraction prediction," Journal of Electronics and Electrical Engineering, vol. 3, no. 1, 2024.

[18] Sarina Thomas, Andrew Gilbert, and Guy Ben-Yosef, "Lightweight spatio-temporal graphs for segmentation and ejection fraction prediction in cardiac ultrasound," in Medical Image Computing and Computer Assisted Intervention – MICCAI 2022. 2022, Springer.

[19] A. Nibali et al., "Numerical coordinate regression with convolutional neural networks," arXiv:1801.07372, 2018.

[20] Yuan-Sen Ting, "Why machine learning models systematically underestimate extreme values," 2024, arXiv:2412.05806.

[21] Jing Lei et al., "Distribution-free predictive inference for regression," Journal of the American Statistical Association, vol. 113, no. 523, pp. 1094–1111, 2018.

[22] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in NeurIPS, 2019.

[23] A. N. Angelopoulos and S. Bates, "Conformal prediction: a gentle introduction," Found. Trends Mach. Learn., vol. 16, no. 4, pp. 494–591, 2023.

[24] Liang-Chieh Chen et al., "Rethinking atrous convolution for semantic image segmentation," arXiv:1706.05587, 2017.

[25] Fatemeh Taheri Dezaki et al., "Cardiac phase detection in echocardiograms with densely gated recurrent neural networks and global extrema loss," IEEE Trans. Med. Imaging, vol. 38, no. 8, pp. 1821–1832, 2019.

[26] Elisabeth S. Lane et al., "Multibeat echocardiographic phase detection using deep neural networks," Comput. Biol. Med., vol. 133, pp. 104373, 2021.

[27] Matthew Christensen et al., "Vision–language foundation model for echocardiogram interpretation," Nature Medicine, vol. 30, pp. 1481–1488, 2024.

[28] Sekeun Kim et al., "EchoFM: Foundation model for generalizable echocardiogram analysis," IEEE Transactions on Medical Imaging, vol. 44, no. 10, pp. 4049–4062, 2025.

[29] Fabian Isensee et al., "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," Nature Methods, vol. 18, no. 2, pp. 203–211, 2021.

[30] Erkang Chen, Zhaoyu Cai, and Jian-Huang Lai, "Weakly supervised semantic segmentation of echocardiography videos via multi-level features selection," in Pattern Recognition and Computer Vision, 2022, pp. 388–400.

[31] Fadillah Maani et al., "UniLVSeg: Unified left ventricular segmentation in echocardiograms through weakly and self-supervised training," arXiv:2310.00454, 2023.

[32] Xiaoxian Yang et al., "MSV-Mamba: A multiscale vision Mamba network for echocardiography segmentation," arXiv:2501.07120, 2025.

[33] Yingyu Yang et al., "Latent motion profiling for annotation-free cardiac phase detection in adult and fetal echocardiography videos," in MICCAI, 2025, pp. 316–325.

[34] J. Martin Bland and Douglas G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," The Lancet, vol. 327, no. 8476, pp. 307–310, 1986.

[35] S. Leclerc et al., "Deep learning for segmentation using an open large-scale dataset in 2D echocardiography," IEEE Trans. Med. Imaging, vol. 38, no. 9, pp. 2198–2210, 2019.